

Cloud Posture Scanner

The Assignment

Build a lightweight **cloud posture tool** that connects to an AWS account (you can use your own test account or free-tier resources).

Your tool should:

1. **Discover Resources**
 - List all EC2 instances with: instance ID, type, region, public IP, and associated security groups.
 - List all S3 buckets with: bucket name, region, encryption status, and access policy (public/private).
2. **Run CIS AWS Benchmark Checks** (at least 5). Examples:
 - No S3 buckets publicly accessible.
 - All S3 buckets encrypted.
 - IAM root account has MFA enabled.
 - CloudTrail is enabled.
 - Security groups are not open to 0.0.0.0/0 for SSH or RDP.
3. **Store Results** securely in AWS (DynamoDB or S3).
4. **Expose REST APIs** to retrieve data, e.g.:
 - /instances
 - /buckets
 - /cis-results
5. **Frontend Dashboard**
 - Display EC2 and S3 data in tables.
 - Show CIS check results (pass/fail with evidence).

⚡ **Note:** We don't expect enterprise-grade completeness. Focus on functionality, coding style, and how you tie components together.

Expected Effort

- MVP scope: ~12–18 hours.
- Polished version (better UI, error handling, deployment): ~20–30 hours.

Interview & Demo

You'll demo:

- How discovery works.
- Which CIS checks you implemented.
- Data flow to AWS.
- API and frontend outputs.
- Design decisions + improvements.